

# REDEMPTION CITY SAFETY COMMAND

## Architecture and Design Document

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A role-based incident management platform for residential camps and large communities, connecting residents, command operators, departments, and field responders in one coordinated safety workflow.

|                  |                                                                                     |
|------------------|-------------------------------------------------------------------------------------|
| Document Purpose | Architecture, workflow, system design, and MVP scope                                |
| Primary Users    | Residents / Reporters, Admin Operators, Departments, Responders                     |
| Core Platforms   | Expo React Native mobile app, Next.js admin dashboard, Supabase backend             |
| Current Status   | Connected MVP with incident reporting, assignment, tracking, and responder workflow |

# 1. Executive Overview

Redemption City Safety Command is designed as a practical safety coordination system for large residential camps, communities, estates, schools, faith-based campuses, and event environments. The product combines a mobile incident reporting app with a web-based command dashboard and a role-based responder workflow.

The goal is simple: make incident reporting faster, make response coordination clearer, and make every action traceable from the moment a resident submits a report until the case is resolved.

## 2. Problem Context

In a large environment, incidents rarely follow one clean communication path. A medical issue, electrical fault, fire alert, theft report, facility issue, traffic obstruction, or lost item may be reported to different people through different channels. This can delay response, create confusion, and make it difficult for command operators to know what has been handled.

The platform addresses these operational gaps by providing one structured channel for reporting, verification, assignment, field updates, and final resolution.

- Residents need a simple way to report safety concerns without searching for the right office or phone number.
- Command operators need a live view of incoming reports, severity, location, evidence, and response status.
- Departments need a reliable way to receive cases that belong to their area of responsibility.
- Responders need a mobile workflow for accepting assignments and updating progress from the field.
- The organization needs accountability through logs, timelines, evidence, and status history.

## 3. Solution Summary

Redemption City Safety Command connects the people involved in incident response through a shared digital workflow. Residents submit reports from the mobile app. The command dashboard receives and verifies those reports. The admin assigns each case to the correct department or responder. The responder then updates progress through the mobile responder interface.

| Resident reports | Command reviews | Department assigned | Responder updates | Reporter notified |
|------------------|-----------------|---------------------|-------------------|-------------------|
|------------------|-----------------|---------------------|-------------------|-------------------|

## 4. Operating Model

The system is built around clearly separated responsibilities. This prevents one person from manually managing every action and allows the platform to scale as more departments and responders are added.

| Role                     | Primary Platform | Main Responsibility                           | Access Scope                 |
|--------------------------|------------------|-----------------------------------------------|------------------------------|
| Resident / Reporter      | Mobile app       | Submit and track reports                      | Own reports only             |
| Admin / Command Operator | Web dashboard    | Verify, assign, monitor, close cases          | All incident operations      |
| Department / Responder   | Mobile app       | Accept assignments and update field progress  | Department or assigned cases |
| Super Admin              | Web dashboard    | Manage users, roles, settings, and operations | Full system control          |

## 5. Core Product Modules

### 5.1 Resident / Reporter Mobile App

The resident app is the first point of contact for the public user. It allows a resident to create an account, submit a report, upload evidence, select location information, and follow the report timeline.

- Incident category, title, description, severity, and contact details.
- Evidence upload for photos or supporting media.
- Location capture or pinned location for operational awareness.
- Tracking timeline for Submitted, In Review, Verified, Assigned, In Progress, Resolved, or Escalated states.

## 5.2 Admin Command Dashboard

The admin dashboard functions as the command center. It provides the operator with a live incident queue, case detail view, assignment tools, responder approval requests, evidence review, notification alerts, and performance overview.

- Live incident queue and critical case indicators.
- Case intelligence: category, source, location, reporter, evidence, and timeline.
- Assignment and dispatch controls for department and responder routing.
- Responder and department management with approval workflow.
- Communication and action logs for audit and accountability.

## 5.3 Responder Mobile Workflow

Responders need a field-ready interface rather than a desktop-only workflow. The responder mobile screen allows approved responders to see their assigned cases, accept tasks, update progress, add feedback, and resolve or escalate a report.

- Assigned incident inbox filtered by responder or department.
- Action buttons for Accept, En Route, Arrived, In Progress, Resolved, and Escalated.
- Responder notes and proof fields for field feedback.
- Realtime assignment refresh and in-app alerts while the app is open.

## 5.4 Department Registration and Approval

To avoid unauthorized access, responders cannot simply create an account and immediately access operational reports. During registration, a responder selects the department they belong to. The request is placed in a pending approval state until an admin reviews and approves it.

This protects sensitive incident data and prevents the mixing of responders from different responsible bodies such as Fire, Medical, C.I.E, Security, Facility, Traffic, and Lost & Found.

## 6. System Architecture

The platform follows a lightweight, cloud-backed architecture. The mobile app and admin dashboard both connect to Supabase for authentication, data, storage, realtime updates, and role-based access control.

| Layer          | Technology                     | Purpose                                                                  |
|----------------|--------------------------------|--------------------------------------------------------------------------|
| Mobile Client  | Expo React Native + TypeScript | Resident reporting and responder field workflow                          |
| Web Client     | Next.js + TypeScript           | Admin command dashboard and operations management                        |
| Authentication | Supabase Auth                  | User login, profiles, and role access                                    |
| Database       | Supabase PostgreSQL            | Incidents, assignments, responders, departments, logs, and notifications |
| Storage        | Supabase Storage               | Incident evidence and media files                                        |
| Realtime       | Supabase Realtime              | Live incident, assignment, and status updates                            |
| Hosting        | Vercel / Supabase Cloud        | Public admin dashboard and backend infrastructure                        |

## 7. Data Model Overview

The MVP database is structured to support the full lifecycle of an incident. The key tables include:

| Table                           | Purpose                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------|
| profiles                        | Stores user identity, role, and basic profile details.                                        |
| incidents                       | Stores incident reports submitted by residents.                                               |
| incident_status_updates         | Stores timeline and status history for each case.                                             |
| departments                     | Stores responsible bodies such as Fire, C.I.E, Medical, Security, Facility, and Lost & Found. |
| responders                      | Stores approved responders linked to departments and mobile login profiles.                   |
| incident_assignments            | Stores department/responder assignments, ETA, dispatch state, and field progress.             |
| responder_registration_requests | Stores pending responder signups for admin approval.                                          |
| communication_logs              | Stores message, call, radio, and admin communication actions.                                 |
| admin_activity_logs             | Stores operational actions performed by command users.                                        |
| notification_queue              | Stores system notification records for reporters and responders.                              |
| push_tokens                     | Stores mobile notification tokens for future production push delivery.                        |
| lost_found_reports              | Stores lost and found reports separately from emergency incidents.                            |

## 8. Incident Lifecycle

1. A resident submits an incident report through the mobile app.
2. The report is stored in Supabase and appears in the admin dashboard.
3. The admin opens the case, reviews the details, and moves it into review.
4. The admin assigns the incident to a responsible department and, where applicable, a specific responder.
5. The responder receives the assignment on mobile and accepts or updates progress.
6. Status updates are recorded in the timeline for command visibility and reporter transparency.
7. The case is marked resolved or escalated based on field feedback and admin confirmation.

## 9. Security and Governance

The platform is designed with role separation and approval controls. A resident should not see command tools, and a responder should not see reports outside their department or assignment scope. Admins have broader access because they coordinate the entire safety operation.

- User access is controlled through Supabase authentication and profile roles.
- Responder access requires department selection and admin approval.
- Incident records, assignments, and timelines are stored centrally for traceability.
- Row Level Security policies are used to guide who can read and update operational records.
- Admin action logs and communication logs support auditability.

## 10. Current MVP Status

| Capability                                     | Status                                                          |
|------------------------------------------------|-----------------------------------------------------------------|
| Resident incident reporting                    | Implemented                                                     |
| Evidence upload                                | Implemented                                                     |
| Admin command dashboard                        | Implemented                                                     |
| Incident assignment and dispatch               | Implemented for MVP testing                                     |
| Responder mobile workflow                      | Implemented for MVP testing                                     |
| Responder department registration and approval | Implemented                                                     |
| Timeline and status updates                    | Implemented                                                     |
| Realtime dashboard refresh and alerts          | Implemented with fallback refresh                               |
| Closed-app push notification delivery          | Prepared; production testing requires development build/APK     |
| Deployment                                     | Admin dashboard prepared for Vercel; backend hosted on Supabase |

## 11. Future Expansion

The MVP provides a strong operational foundation. Future versions can expand the product into a deeper emergency management and facility operations platform.

- Production Android APK and iOS build for wider testing.
- Full push notification delivery outside Expo Go.
- Advanced GIS map routing and incident heatmaps.
- Department-specific web portals and responder performance dashboards.
- SMS, WhatsApp, and email communication integrations.
- AI-assisted triage for incident classification and priority recommendation.
- Service level tracking, response time analytics, and escalation rules.
- Offline-first reporting for poor-network environments.

## 12. Impact Statement

Redemption City Safety Command strengthens safety coordination by turning scattered reports into an organized command workflow. It helps residents know that their reports are visible, helps admins understand what is happening in real time, and helps responders act with clearer instructions.

The result is a safer, more accountable operating environment where reports can be tracked, assignments can be monitored, and response history can be reviewed after each incident.